01 · LESSON

Bias, validity and organic farming

9B Plant growth · Lesson 7 of 7

Date _____ Class _____

Today I can

- ☐ I can identify bias in graphs and arguments, explain where accidental bias comes from, and say whether a conclusion is valid.
- ☐ I can compare organic and intensive farming and explain biological control.

What happens today

1. Do Now: monarchs and milkweed
2. Learn: Bias on purpose · Bias by accident · Is it valid? · and more
3. Core tasks
4. Challenge, then the exit ticket

Student Book pages 34 to 36 · Workbook pages 25 to 26

What I learned today (one or two sentences)

02 · KEY VOCABULARY

Key Vocabulary

Term	Definition
bias	a shift away from the true meaning or value, on purpose or by mistake
balanced	considering the points for and against something equally
systematic error	an error that shifts every reading by the same amount, e.g. a wrongly printed scale
random error	an error that changes from one reading to the next, e.g. stopping a clock late once
sample	a small part of a whole that is measured to estimate the whole
valid	does what it is supposed to do; a valid conclusion uses only the results
conclusion	what the results of an investigation show
organic farming	farming with natural fertilisers, very few pesticides and crop rotation
intensive farming	farming with artificial fertilisers, pesticides and large fields to get the biggest yield
biological control	using living organisms to reduce the number of pests
crop rotation	growing different crops in a field in different years

Quick Review

monarchs and milkweed

Monarch butterflies lay their eggs only on milkweed plants. HT soybean is a variety that is not killed by the herbicide glyphosate, which kills milkweed.

1 Predict what happens to milkweed where HT soybean is grown.

2 Predict the effect on monarch butterflies.

3 Suggest one other reason why monarch numbers could fall.

Core 1 of 3

Everyone does these. Answer on the card or under it.

1 Students added different numbers of drops of liquid fertiliser to bean seedlings and measured the mean increase in height after a week.

Number of drops of liquid fertiliser	2	4	6	8	10
Mean increase in height of seedlings (cm)	4.6	6.3	7.8	7.5	7.2

a) Tick one box to show a valid conclusion for this experiment.

- ☐ A Seedling height increases as the amount of fertiliser increases.
- ☐ B Too little or too much fertiliser can harm seedlings.
- ☐ C The more fertiliser that is added, the smaller the increase in growth.
- ☐ D Growth increases with more fertiliser up to a point, beyond which the amount of growth falls.

b) Give a reason for your choice.

Core 2 of 3

Everyone does these. Answer on the card or under it.

1 *continued*

c) State two variables that should be controlled to make this experiment valid.

2 Match each scientific term to the correct example. Draw a line.

systematic error	the stop clock was stopped late in one test during an experiment
random error	only data that support a particular view are quoted
valid experiment	a mass balance is not set to zero before the experiment
intentional bias	all other variables are controlled, so the dependent variable is only affected by the independent variable

Core 3 of 3

Everyone does these. Answer on the card or under it.

3 Tick one box in each row to show whether the statement describes organic farming or intensive farming.

Statement	Organic	Intensive
mostly natural fertilisers such as manure		
many artificial pesticides		
large fields to make it easy to use big machines		
small fields with hedges and trees for predators of pests to live in		
crops are rotated so that clover adds nitrates to the soil		
the biggest yield from each field		

Challenge 1 of 2

When Core is finished. Full sentences.

C1 HT soybean is not killed by the herbicide glyphosate. Advantages: easier management of weeds; fewer sprayings needed than with other herbicides; glyphosate is not persistent. Disadvantages: expensive seed; with time, weeds unaffected by glyphosate become more common, so other herbicides may be needed; HT soybean may grow as a weed in other crops. Write a balanced paragraph to help a farmer decide whether to plant HT soybean.

Challenge 2 of 2

When Core is finished. Full sentences.

C2 Two farms have the same area. One is organic and one is intensive.

a) Which farm produces the greater amount of crop? Explain your answer.

b) Explain why some people still choose to buy organic food, and state what the evidence says about whether organic food is healthier.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 What is bias?

2 Give one example of a systematic error.

3 Give one difference between organic farming and intensive farming.

Today I can: Goal 1 ☐ Goal 2 ☐

09 · EXIT TICKET

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 What is bias?

2 Give one example of a systematic error.

3 Give one difference between organic farming and intensive farming.

Today I can: Goal 1 ☐ Goal 2 ☐

09 / 09 9B L7 · Bias, validity and organic farming EXIT